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1. J1/R1/R2/U1/F2 USB-C 5 V charge input: UFP 5.1k on CC1/CC2, USBLC6-2SC6 ESD, polyfuse
2. U2/R3..R8/TH1 TP4056 1 A charger (PROG=1.2k), NTC temperature window, ADC dividers
3. BT1/BT2/J2/J3/F3/SW1 battery pack 1S2P + BMS 1S2P (B-/B+/P-/P+) + polyfuse + OFF switch
4. U3/L1/D2/R12/R13 MT3608 boost 12.22 V (Vout=0.6*(1+91k/4.7k), sense AFTER D2)
5. J4/F1/D3/D4/C9 external 12 V input with SS34 OR-ing + TVS; priority = external
6. J5/J6/TP1..TP3 12 V OUT to CORE-1; BAT_ADC/TEMP_ADC to MCP3208 CH4/CH5; test points

Auto source: external 12 V -> EXT_DET (R10/R11) -> Q1 pulls MT3608 EN low -> boost off,
battery not drained. External removed: Q1 off, R9 pulls EN up, boost restarts at 12.22 V.
GND here is one net; on CORE-1 it lands on the PGND/AGND star point (NT1/NT2).

Bukifon open hardware - CERN-OHL-S-2.0

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Title: PWR-BATT-1 — аккумуляторный блок питания Bukifon (1S2P Li-ion +

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